Does Underlying Structure Predict Quantum Trainability and Dequantization?

Charbel A. Dargham · 25-line summary

We ask whether the underlying structure of a learning problem (symmetry, locality, spectrum and data complexity) can predict, before training, whether structure-preserving quantum models of it will be trainable and resist dequantization. Random circuits suffer barren plateaus (McClean et al. 2018); building the problem's symmetry into the circuit keeps gradients large through a small dynamical Lie algebra (Larocca et al. 2022; Ragone et al. 2024), but the same smallness can let a classical computer imitate the model (Cerezo et al. 2025; Goh et al. 2023). Only symmetry has a theorem linking the problem to the trainability formula; for the other three kinds the link is our working hypothesis. Phase A builds problems, circuits derived from their structure, fingerprints and measurements; Phase B learns a predictor tested on families it has never seen. This study validates the first links: the dataset and the structure.

A problem is one binary learning task at fixed n (4 to 12 qubits), $\mathscr{P}_n = (\mathscr{X}_n, \mathscr{Y}_n, \mu_n, \mathscr{E}_n, f^*)$, where each input determines a differential equation through an operator $M(x)$, and the operator fixes the type: GRAPH (Laplacian), COVARIANCE (precision matrix), DIAGONAL (clause-penalty Hamiltonian) and QUANTUM (non-commuting Hamiltonian). The audit of the raw harvest (1 379 problems) found TRAIN/TEST leaks, one unbalanced problem, degenerate ground states, non-converged molecules and uninformative operators. The raw data were never modified; six stated rules produced the final dataset: 1 368 problems, 58 source groups, 337 930 samples, 0 audit failures and 0 failed equation checks.

Each operator becomes a factor graph carrying 12 scale-free features. Q2 (mixed models with a random effect per source): the spectral features differ between types and scale differently with n, but only GRAPH separates, while COVARIANCE, DIAGONAL and QUANTUM stay close; ICC 0.5–0.9 leaves 18–132 effective observations. Q3 (classifier, one source held out at a time): types are recovered at 0.91 balanced accuracy, errors fall where the operator definitions predict, but domains within a type are not. Q4: features change monotonically with n, all 28 forced laws hold exactly and critical XXZ chains follow Calabrese–Cardy; only 19 of 177 laws are nameable at $n \leq 12$.

The extracted structure is correct, varied beyond construction yet coarse, and lawful in n: a predictor built on it will see kinds of operators rather than domains, and describe finite sizes rather than asymptotics. Next: permutation tests, re-harvesting the 10 XY-periodic problems, Hodge features, then circuits and measurements validated on held-out sources.